Physics 9702 Paper 4 · Topic 18

Electric fields

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

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1 9702/41 October/November 2020, Question 5 (9 marks) page 2

Examiner report: (b) (i) Candidates found determining this charge very challenging.

2 9702/41 October/November 2022, Question 4 (12 marks) page 4

Examiner report: (c) (i) Candidates found drawing the path of the proton challenging. Some candidates realised that the charge- to-mass ratio was important in this question but very few candidates separated the velocities into their vertical and horizontal components.

3 9702/42 October/November 2022, Question 5 (10 marks) page 6

Examiner report: Comparatively few candidates appreciated that the negative nature of the potential curve meant that the sign of the charge on the sphere must be negative.

4 9702/41 October/November 2023, Question 5 (8 marks) page 8

Examiner report: (c) This was the most challenging part of the question and one of the most difficult questions on the question paper.

5 9702/41 May/June 2022, Question 2 (8 marks) page 10

Examiner report: Candidates found this question challenging as it involved gravitational fields, electric fields and magnetic fields. (a) (i) Candidates found it difficult to determine the correct direction for the magnetic field.

6 9702/42 February/March 2023, Question 4 (12 marks) page 12

Examiner report: (ii) This multistage calculation proved to be challenging to most candidates. However, only the strongest candidates were able to resolve forces and apply the equilibrium conditions.

7 9702/44 October/November 2025, Question 5 (9 marks) page 15

Examiner report: (a) Very few candidates distinguished between ‘work done on the charges’ and ‘work done by the charges’. Many candidates stated that the force was repulsive without explaining that this was because the charge on a proton is positive.

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

5 **(a)** Define *electric potential* at a point.

.....

.....

..... [2]

(b) Two point charges A and B are separated by a distance of 12.0 cm in a vacuum, as illustrated in Fig. 5.1.

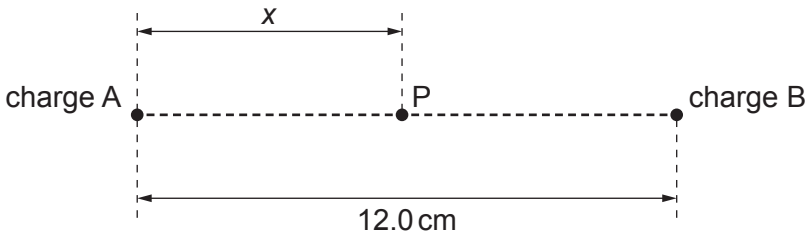


Fig. 5.1

The charge of A is $+2.0 \times 10^{-9}$ C.

A point P lies on the line joining charges A and B. Its distance from charge A is x.

The variation with distance x of the electric potential V at point P is shown in Fig. 5.2.

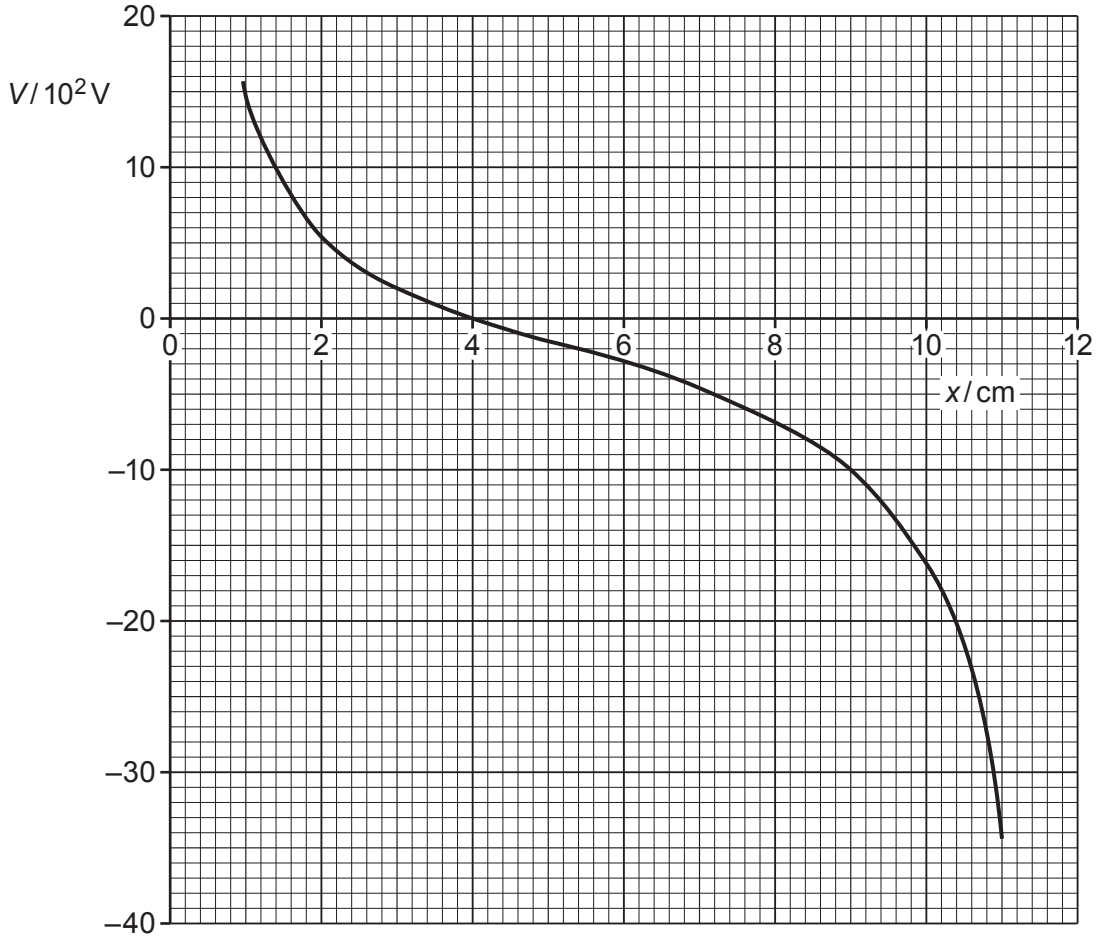


Fig. 5.2

Use Fig. 5.2 to determine:

(i) the charge of B

charge = ... C [3]

(ii) the change in electric potential when point P moves from the position where $x = 9.0\,\text{cm}$ to the position where $x = 3.0\,\text{cm}$.

change = ... V [1]

(c) An α-particle moves along the line joining point charges A and B in Fig. 5.1.

The α-particle moves from the position where $x = 9.0\,\text{cm}$ and just reaches the position where $x = 3.0\,\text{cm}$.

Use your answer in **(b)(ii)** to calculate the speed v of the α-particle at the position where $x = 9.0\,\text{cm}$.

v = ... m s^{-1} [3]

[Total: 9]

4 **(a)** State what is indicated by the direction of an electric field line.

.....

..... [2]

(b) Fig. 4.1 shows a pair of parallel metal plates with a potential difference (p.d.) of 2400 V between them.

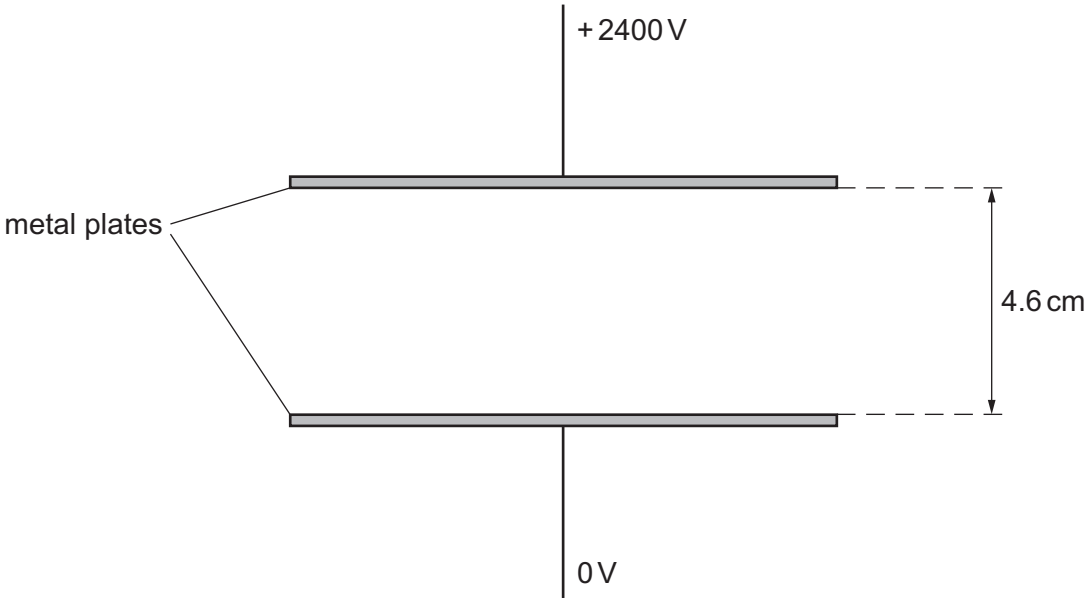


Fig. 4.1

The plates are separated by a distance of 4.6 cm. The plates are in a vacuum.

(i) On Fig. 4.1, draw five lines to represent the electric field in the region between the plates. [3]

(ii) Calculate the strength of the electric field between the plates.

electric field strength = ... $N C^{-1}$ [2]

(c) A moving proton enters the region between the plates from the left, as shown in Fig. 4.2.

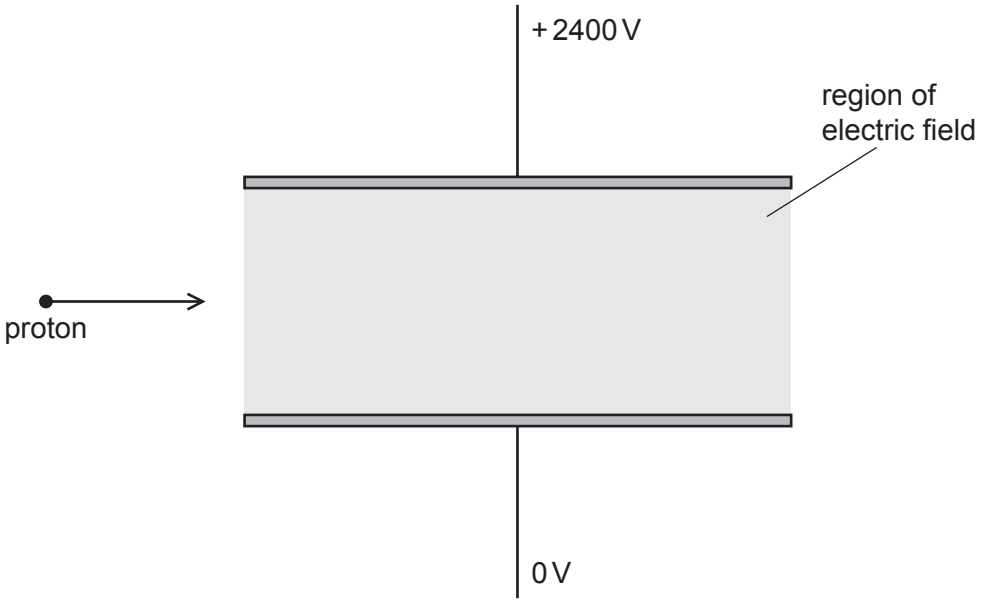


Fig. 4.2

(i) The proton is deflected by the electric field.

On Fig. 4.2, draw a line to show the path of the proton as it moves through and out of the region of the electric field. [2]

(ii) A helium nucleus ($^4_2$He) now enters the region of the electric field along the same initial path as the proton and travelling at the same initial speed.

State and explain how the final speed of the helium nucleus compares with the final speed of the proton after leaving the region of the electric field.

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.....

.....

..... [3]

[Total: 12]

5 **(a)** Define electric potential at a point.

.....

.....

..... [2]

(b) An isolated conducting sphere is charged. Fig. 5.1 shows the variation of the potential V due to the sphere with displacement x from its centre.

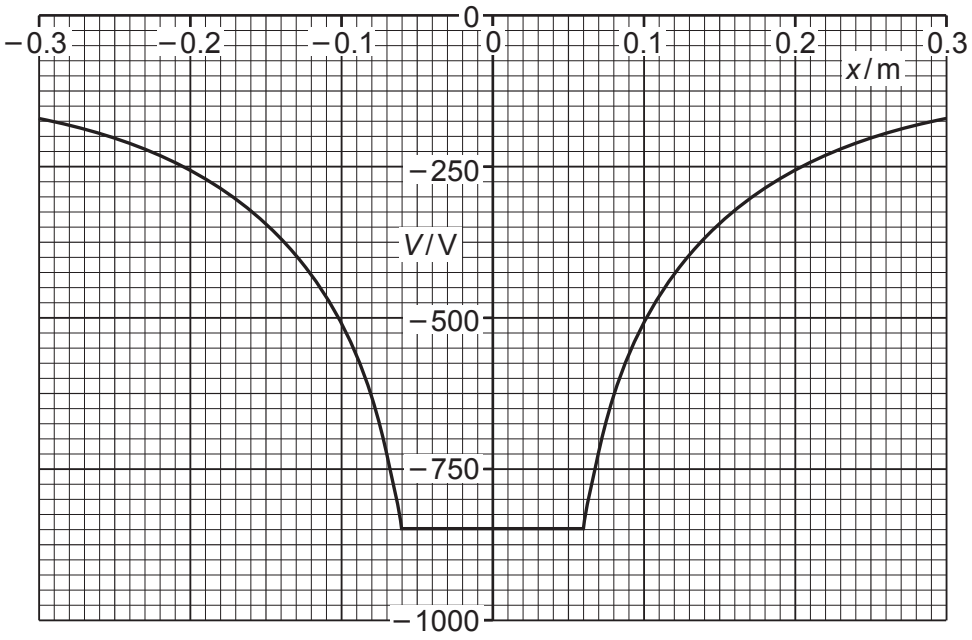


Fig. 5.1

Use Fig. 5.1 to determine:

(i) the radius of the sphere

radius = ... m [1]

(ii) the charge on the sphere.

charge = ... C [2]

(c) Two spheres are identical to the sphere in **(b)**. Each sphere has the same charge as the sphere in **(b)**.

The spheres are held in a vacuum so that their centres are separated by a distance of 0.46 m. Assume that the charge on each sphere is a point charge at the centre of the sphere.

(i) Calculate the electric potential energy E_P of the two spheres.

E_P = ... J [2]

(ii) The two spheres are now released simultaneously so that they are free to move.

Describe and explain the subsequent motion of the spheres.

.....

.....

.....

..... [3]

[Total: 10]

5 **(a)** Define electric potential at a point.

.....

.....

..... [2]

(b) Two isolated charged metal spheres X and Y are situated near to each other in a vacuum with their centres a distance of 24 m apart. Point P is at a variable distance *x* from the centre of sphere X on the line joining the centres of the spheres.

Fig. 5.1 shows the variation with *x* of the electric potential *V* due to the spheres at point P.

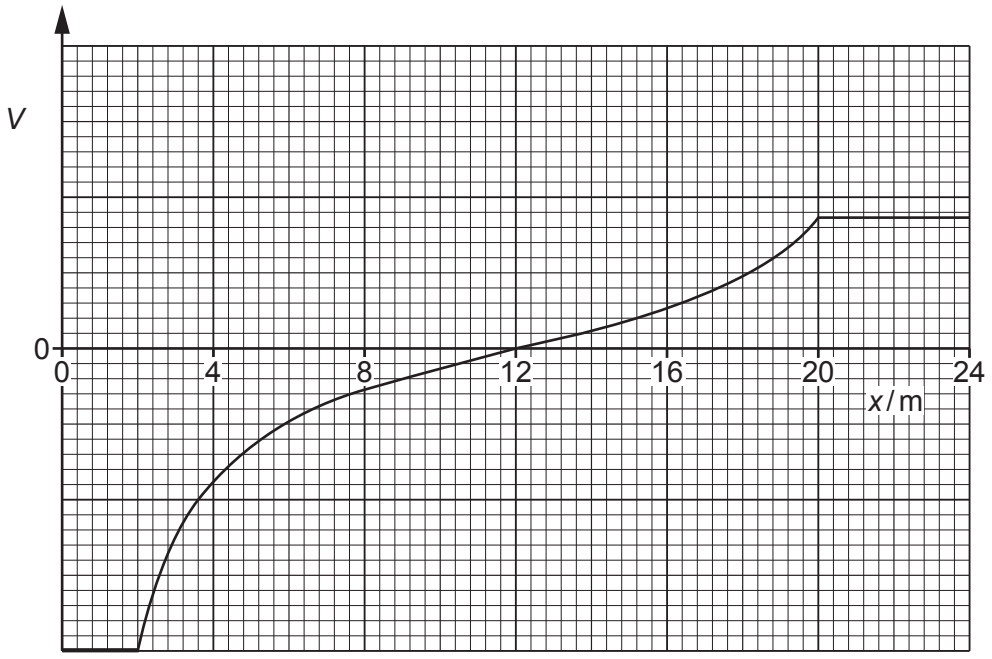


Fig. 5.1

State **three** conclusions that can be drawn about the spheres from Fig. 5.1. The conclusions may be qualitative or quantitative.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) A positively charged particle is placed at point P in **(b)**, such that $x = 12\,\text{m}$.
The particle is released.

Describe and explain the subsequent motion of the particle.

.....

.....

.....

.....

..... [3]

[Total: 8]

2 A sphere of mass 1.6×10^{-10} kg has a charge of +0.27 nC. The sphere is in a uniform electric field that acts vertically upwards, as shown in the side view in Fig. 2.1.

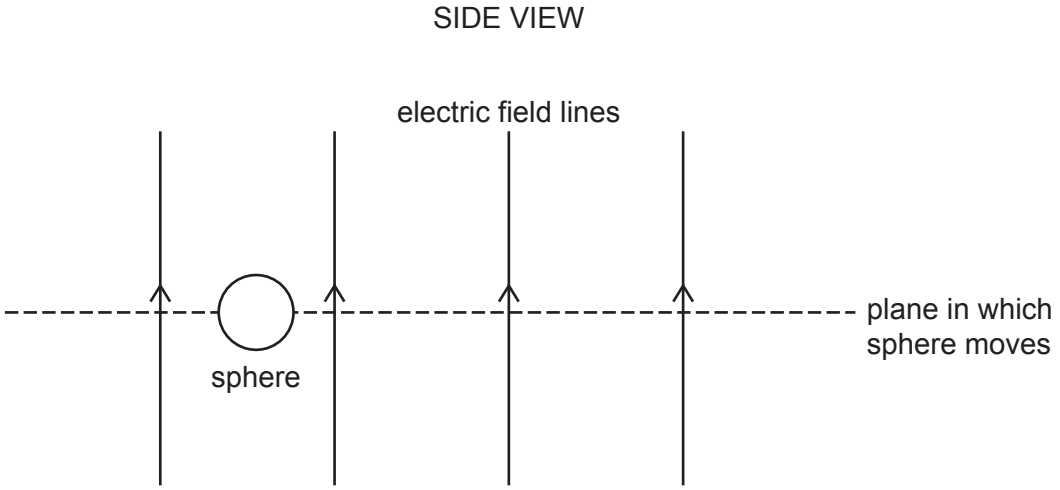


Fig. 2.1

The force exerted on the sphere by the electric field causes the sphere to remain at a fixed vertical height in a horizontal plane.

There is a uniform magnetic field in the region of the electric field. The sphere moves at a speed of $0.78\,m\,s^{-1}$ in the horizontal plane. The magnetic field causes the sphere to move in a circular path of radius 3.4 m, as shown in the view from above in Fig. 2.2.

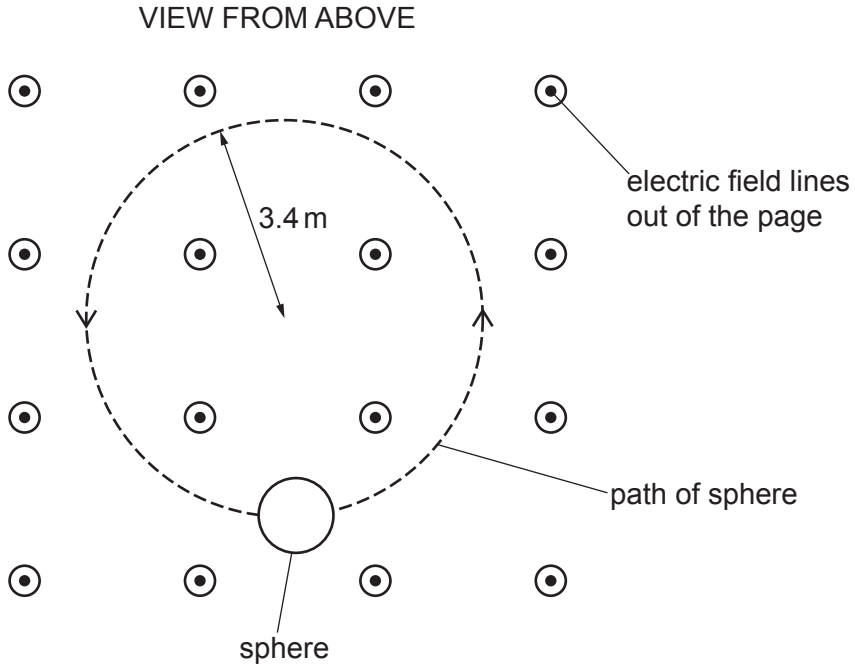


Fig. 2.2

(a) **(i)** Determine the direction of the uniform magnetic field.

..... [1]

(ii) Explain why the motion of the sphere in the horizontal plane is circular.

.....

.....

..... [2]

(b) Calculate the strength of the uniform electric field.

electric field strength = ... $N\,C^{-1}$ [2]

(c) By considering the magnetic force on the sphere, show that the flux density of the uniform magnetic field is 0.14 T.

[3]

[Total: 8]

4 **(a)** State Coulomb's law.

.....

.....

.....

..... [2]

(b) A charged sphere X is supported on an insulating stand. A second charged sphere Y is suspended by an insulating thread so that sphere Y is in equilibrium at the position shown in Fig. 4.1.

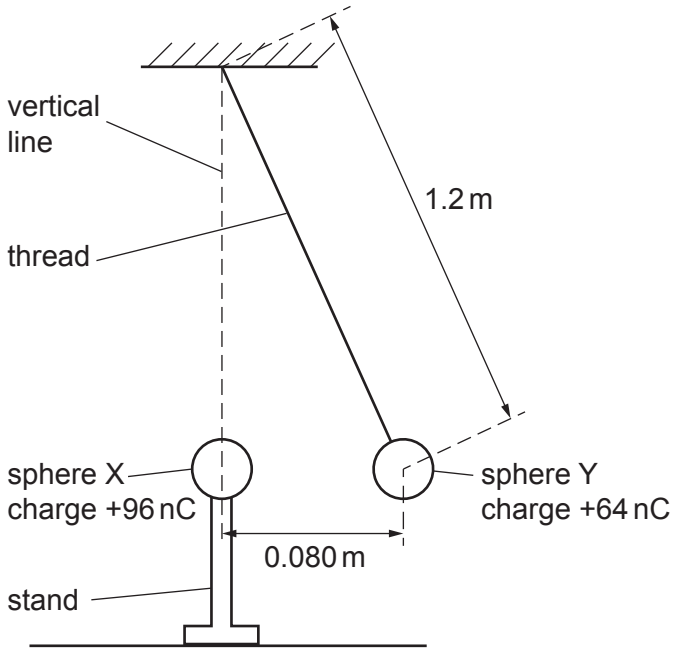


Fig. 4.1

The charge on sphere X is +96 nC and the charge on sphere Y is +64 nC.
Assume that the spheres behave as point charges.

The length of the thread is 1.2 m and the centres of sphere X and sphere Y are separated horizontally by a distance of 0.080 m.

(i) On Fig. 4.2, draw and label all the forces acting on sphere Y.

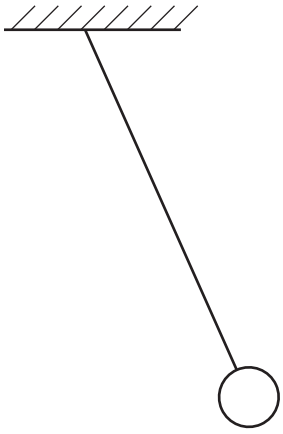


Fig. 4.2

[1]

(ii) Determine the mass of sphere Y.

mass = ... kg [4]

(iii) Calculate the total electric potential energy stored between X and Y.

energy = ... J [1]

(c) An electron enters the region between two parallel plates P and Q, that are separated by a distance of 18 mm, as shown in Fig. 4.3.

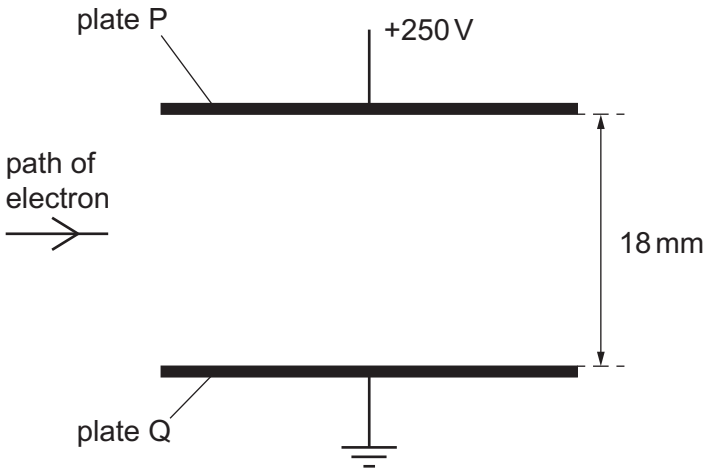


Fig. 4.3

The space between the plates is a vacuum.
The potential difference between the plates is 250 V. The electric field may be assumed to be uniform in the region between the plates and zero outside this region.

(i) State the direction of the electric force on the electron when between the plates.

..... [1]

(ii) Determine the magnitude of the force acting on the electron due to the electric field.

force = ... N [2]

(iii) Explain why the electron does **not** follow a circular path.

.....

..... [1]

[Total: 12]

5 **(a)** Explain why the electric potential near an isolated proton is positive.

.....

.....

.....

.....

..... [3]

(b) An isolated metal sphere is positively charged and has radius *R*, as shown in Fig. 5.1.

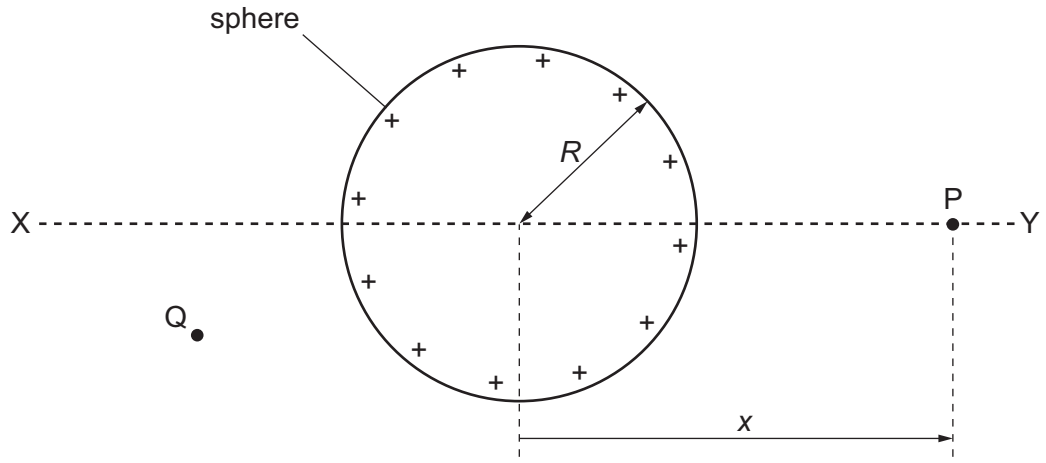


Fig. 5.1

Line XY passes through the centre of the sphere.
Point P lies on line XY at a variable displacement *x* from the centre of the sphere.
Point Q is at a fixed position that is not on line XY.

The electric field strength at the surface of the sphere is E_0.

(i) On Fig. 5.1, draw an arrow at point Q to show the direction of the electric field at that point. [1]

(ii) On Fig. 5.2, sketch the variation of the electric field E at point P with x for values of x between $x = -3R$ and $x = 3R$. Do **not** include the region inside the sphere between $x = -R$ and $x = R$.

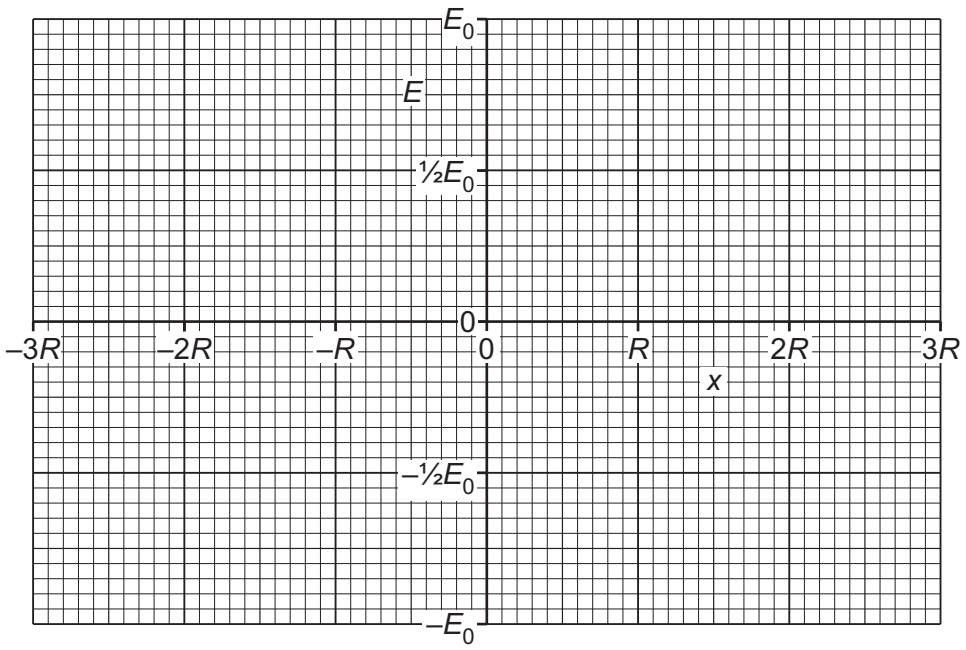


Fig. 5.2

[3]

(c) The proton and the electron in a hydrogen atom are separated by a distance of 5.3×10^{-11} m.

Calculate the electric potential energy of the proton and the electron.

electric potential energy = ... J [2]

[Total: 9]

Question	Answer	Marks
5(a)	work done per unit charge	**B1**
	(work done on charge) moving positive charge from infinity	**B1**
5(b)(i)	$(2.0 \times 10^{-9}) / 4\pi\varepsilon_0(4.0 \times 10^{-2}) + Q / 4\pi\varepsilon_0(8.0 \times 10^{-2}) = 0$	**C1**
	$Q = 4.0 \times 10^{-9}$ C	**A1**
	Q given with negative sign	**B1**
5(b)(ii)	change = 1200 V	**A1**
5(c)	$\frac{1}{2}mv^2 = qV$	**C1**
	$\frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2 = 2 \times 1.60 \times 10^{-19} \times 1200$	**C1**
	$v = 3.4 \times 10^5$ m s^{-1}	**A1**

Question	Answer	Marks
4(a)	(field line indicates) direction of force	**B1**
	force on a positive charge	**B1**
4(b)(i)	one straight line perpendicular to plates, starting on one plate and finishing on the other	**B1**
	five straight lines perpendicular to plates between the plates, uniformly spaced	**B1**
	downwards arrows on lines	**B1**
4(b)(ii)	$E = V / d$	**C1**
	$= 2400 / 0.046$ $= 5.2 \times 10^4\ \text{N C}^{-1}$	**A1**
4(c)(i)	smooth curve in region of field and straight line outside field	**B1**
	direction of deflection shown as downwards in region of field	**B1**
4(c)(ii)	helium nucleus has double the charge but four times the mass	**B1**
	velocity parallel to plates same and acceleration perpendicular to plates smaller (for helium)	**B1**
	final speed is lower (for helium)	**B1**

Question	Answer	Marks
5(a)	work done per unit charge	**B1**
	work done (on charge) in moving positive charge from infinity (to the point)	**B1**
5(b)(i)	radius = 0.060 m	**A1**
5(b)(ii)	$V = Q / 4\pi\varepsilon_0 x$ Q = (–) 850 × 4π × 8.85 × 10^{-12} × 0.060 **or** Q = (–) 850 × 0.060 / 8.99 × 10^{9} (*any correct pair of V and x values from curve*)	**C1**
	Q = – 5.7 × 10^{-9} C	**A1**
5(c)(i)	$E_P = Q^2 / 4\pi\varepsilon_0 x$ = (5.67 × 10^{-9})^2 / (4π × 8.85 × 10^{-12} × 0.46)	**C1**
	= 6.3 × 10^{-7} J	**A1**
5(c)(ii)	• force is repulsive so spheres move apart • force in direction of motion so speed increases • potential energy converted to kinetic energy so speed increases • force decreases with distance so acceleration decreases • momentum is conserved (at zero) (and masses are equal) so velocities are always equal and opposite *Any three points, 1 mark each*	**B3**

Question	Answer	Marks
5(a)	work done per unit charge	**B1**
	work (done on charge) moving positive charge from infinity (to the point)	**B1**
5(b)	*Any three points from*: *Up to 2 points from*: • radius of sphere X is 2.0 m • radius of sphere Y is 4.0 m • radius of Y is double the radius of X *Up to 2 points from*: • charge on X is negative • charge on Y is positive • spheres carry opposite charges *Up to 1 point from*: • magnitudes of charges on the spheres are equal	**B3**
5(c)	particle is attracted to X or repelled from Y **or** resultant force on particle is towards X / away from Y / to the left	**B1**
	particle accelerates towards X / away from Y / to the left	**B1**
	(magnitude of) acceleration of particle increases	**B1**

Question	Answer	Marks
2(a)(i)	(vertically) downwards	**B1**
2(a)(ii)	magnetic force (on sphere) is perpendicular to its velocity	**B1**
	magnetic force perpendicular to velocity is the centripetal force **or** magnetic force perpendicular to velocity causes centripetal acceleration **or** acceleration perpendicular to velocity is centripetal (acceleration) **or** magnetic force does not change the speed of the sphere **or** magnetic force has constant magnitude	**B1**
2(b)	$mg = Eq$	**C1**
	$E = (1.6 \times 10^{-10} \times 9.81) / (0.27 \times 10^{-9})$ $= 5.8\ \text{N C}^{-1}$	**A1**
2(c)	centripetal force = magnetic force **or** $Bqv = mv^2 / r$	**B1**
	$B = mv / qr$	**C1**
	$= (1.6 \times 10^{-10} \times 0.78) / (0.27 \times 10^{-9} \times 3.4) = 0.14\ \text{T}$	**A1**

Question	Answer	Marks
4(a)	(electric) force is (directly) proportional to product of charges	**B1**
	force (between point charges) is inversely proportional to the square of their separation	**B1**
4(b)(i)	arrows showing tension upwards in direction of string, electric force horizontally to the right and weight vertically downwards and all three labelled	**B1**
4(b)(ii)	$F_E = \dfrac{96\times10^{-9}\times64\times10^{-9}}{4\times\pi\times8.85\times10^{-12}\times0.080^2}$ $(= 8.63\times10^{-3}\,\text{N})$	**C1**
	either angle to vertical $= \sin^{-1} 0.080 / 1.2$ $(= 3.82°)$	**C1**
	weight $= F_E / \tan 3.82 = 8.63\times10^{-3} / \tan 3.82$ $(= 0.129\,\text{N})$	**C1**
	mass $= 0.129 / 9.81$ $= 0.013\,\text{kg}$	**A1**
	or $T\sin\theta = mg$ **and** $T\cos\theta = F_E$ **or** $\tan\theta = mg / F_E$	**(C1)**
	$\tan\theta = 1.2 / 0.080$	**(C1)**
	$m = (1.2\times8.63\times10^{-3}) / (0.080\times9.81)$ $= 0.013\,\text{kg}$	**(A1)**

Question	Answer	Marks
4(b)(iii)	$E_p = \frac{Q_1 Q_2}{4\pi\varepsilon_o r} = \frac{96\times10^{-9}\times64\times10^{-9}}{4\times\pi\times8.85\times10^{-12}\times0.080}$ $= 6.9 \times 10^{-4}$ J	**A1**
4(c)(i)	towards the top of the page / towards plate P	**B1**
4(c)(ii)	$F = QE$ **and** $E = V / d$	**C1**
	$F = 1.6 \times 10^{-19} \times 250 / 0.018$ $= 2.2 \times 10^{-15}$ N	**A1**
4(c)(iii)	**either** the force is not (always) perpendicular to the velocity **or** the force is always in the same direction	**B1**

Question	Answer	Marks
5(a)	potential is (defined as) zero at infinity	**B1**
	proton has a positive charge and so repels another positive charge	**B1**
	work is done on two (positive) charges to move them towards each other **or** work is done by two (positive) charges as they move apart from each other	**B1**
5(b)(i)	arrow drawn through Q in a WSW direction directly away from the centre of the sphere	**B1**
5(b)(ii)	curve in at least one quadrant passing through ($\pm R$, $\pm E_0$) and ($\pm 2R$, $\pm ¼E_0$)	**B1**
	curve between $-3R$ and $-R$ of increasing magnitude of gradient **and** curve between R and $3R$ of decreasing magnitude of gradient	**B1**
	two lines drawn, one in the top right quadrant, the other in the bottom left quadrant	**B1**
5(c)	$E_P = -(1.60 \times 10^{-19})^2 / [4\pi \times 8.85 \times 10^{-12} \times (5.3 \times 10^{-11})]$	**C1**
	$= -4.3 \times 10^{-18}$ J	**A1**